How to handle LLMs in your research

Pablo Mosteiro, Robert Bagheri, Özgür Togay, Laurence Frank

Partly generated with help of Qwen/Qwen2.5-7B-Instruct and moonshotai/Kimi-K2-Instruct-0905 via together and groq, respectively (HuggingChat).

Department of Methodology, Statistics & Data Science

September 2, 2026

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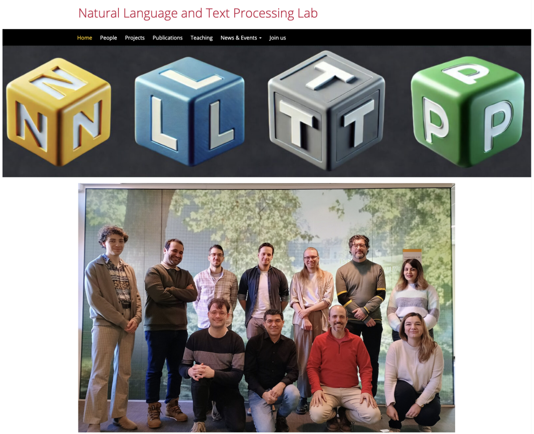
Today's Agenda

- Introduction (3 min)
- How LLMs Work (10 min)
- Limitations and Risks (10 min)
- Prompting CAs and Chatbots (10 min)
- Practical LLM Use in Research (10 min)
- Conclusion (2 min)

About us



https://nlp.sites.uu.nl/

Introduction: What do you know about LLMs?

- **Activity: Think-Pair-Share** (2 min)

Image produced in DALL-E December 2023 (Credit: Christine Fox)

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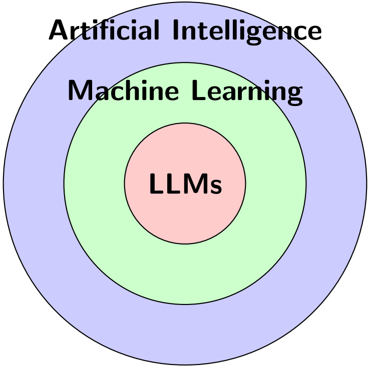
AI vs ML vs LLM: The Relationship



LLMs are a specific type of ML,
which is a subset of AI

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Model Learning Process: Training Phase

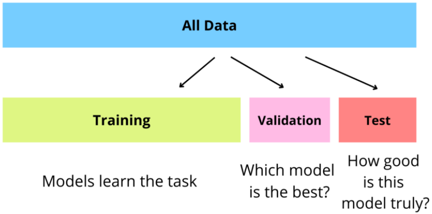
Training Phase

- Model learns patterns from large datasets
- Adjusts model to minimize prediction errors
- Requires massive computational resources
- Can take weeks or months for large models

Example: LLM learns that "cat" often appears with "dog" in pet-related contexts

Model Learning Process: Validation Phase

Validation and Testing



Source: `https://dqlab.id/5-panduan-praktis-membuat-model-machine-learning`

Key insight: Good models **generalize**, not just **memorize**

Prompting phase

U weet wat ik ga zeggen

Credit for this slide: Albert Gatt & Dong Nguyen

Prompting phase

U weet wat ik ga

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What is Generative AI?

Definition: AI that creates new content (text, images, code, audio)
Key Features:

- Creates novel outputs, not just classifies
- Learns patterns from training data
- Generates human-like content

Examples: ChatGPT (text), DALL-E (images), CodeT5 (code)

Text-GenAI components

consists of	fed by	GPT learns
LLM	textual internet data	"knowledge" of language and content
Dialogue Model	social media, forums	"knowledge" of dialogue structure
Task Model	programmers	tasks, self knowledge and limits
Ethical Governor	Kenian data labellers	ethical answers

Credit for this slide: Davitze Könning

Generative AI vs. Other Models

Feature	Generative	Discriminative
Primary Task	?	?
Output Type	?	?
Example	?	?

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Generative AI vs. Other Models

Feature	Generative	Discriminative
Primary Task	Create new content	Classify existing data
Output Type	Novel text/images	Labels/categories
Example	GPT, DALL-E	Spam filters, medical diagnosis

Activity: True or False?

Instructions: Raise your hand if you think the statement is TRUE

- **Statement 1**: Generative AI can only work with text
- **Statement 2**: All generative models are also discriminative

Caveat

gold labels!

- **Statement 3**: Generative AI creates completely original content never seen in training
- **Statement 4**: LLMs are a type of generative AI

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Limitations and Risks: Hallucinations

What are hallucinations?

- LLMs generate false or nonsensical information
- Appears confident and plausible
- Can include fake citations, non-existent facts

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Bias

- **Gender bias:** Assumes doctors are male, nurses are female
- **Racial bias:** Associates certain groups with negative stereotypes
- **Cultural bias:** Primarily represents Western perspectives
- **Academic bias:** Favors published, English-language sources
- **Confirmation bias:** Reinforces existing beliefs
- **Temporal bias:** Becomes outdated as language/society changes

Real-world impact: Biased models can perpetuate harmful stereotypes

Limitations and Risks: Output Reliability

Reliability Issues

- **Inconsistency:** Same prompt can yield different answers

Give me the answer to the question "What is life?"

The question "What is life?" is a profound and complex one that has been explored by philosophers, scientists, theologians, and thinkers throughout history. There isn't a single, universally accepted answer, but here are some perspectives:

Give me the answer to the question "What is life?"

The question "What is life?" is a profound and complex one that has been explored by various disciplines, including biology, philosophy, theology, and more. Here are some key perspectives:

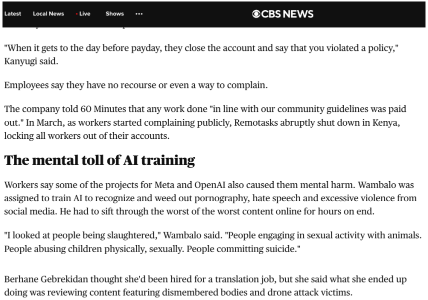
- **Context dependency:** Performance varies by domain

Pre-Chorus:

Limitations and Risks: Ethics

Ethical Concerns

- **Data:** Training data may include personal or copyrighted material
- **Environmental impact:** High energy consumption
- **Labor practices:** Data annotation often involves poor working conditions



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"When it gets to the day before payday, they close the account and say that you violated a policy," Kanyugi said.

Employees say they have no recourse or even a way to complain.

The company told 60 Minutes that any work done "in line with our community guidelines was paid out." In March, as workers started complaining publicly, Remotasks abruptly shut down in Kenya, locking all workers out of their accounts.

The mental toll of AI training

Workers say some of the projects for Meta and OpenAI also caused them mental harm. Wambalo was assigned to train AI to recognize and weed out pornography, hate speech and excessive violence from social media. He had to sift through the worst of the worst content online for hours on end.

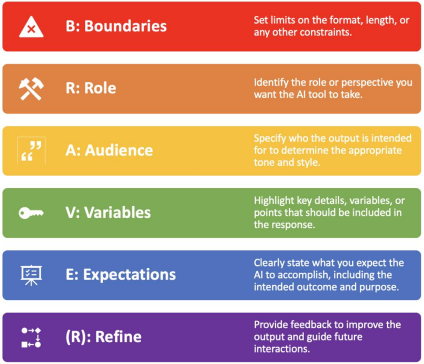
"I looked at people being slaughtered," Wambalo said. "People engaging in sexual activity with animals. People abusing children physically, sexually. People committing suicide."

Berhane Gebrekidan thought she'd been hired for a translation job, but she said what she ended up doing was reviewing content featuring dismembered bodies and drone attack victims.

`https://www.cbsnews.com/news/ai-work-kenya-exploitation-60-minutes` (Nov 2024)

- **Academic integrity:** attribution of AI assistance / uploading restricted data
- **Open source:** Often little documentation on data, training, architecture

Effective Prompting: BRAVE(R) Framework



Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).

Poor prompt: "Summarize this paper"

Better prompt: "Act as a psychology research assistant [R]. Summarize this paper in 3 bullet points [B] focusing on methodology and findings relevant to cognitive behavioral therapy [V]. Use academic language [A] and APA style [E]."

Key insight: Specific, contextual prompts yield better results

Evaluating and Critically Assessing GenAI Responses

Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).

- **F: Focus** — Focus on the AI's response by breaking it down into its main points and arguments.
- **A: Authenticate** — Authenticate the AI's output by verifying it against current research and empirical data.
- **C: Critique** — Critique the response's accuracy, relevance, and depth in the context of the topic.
- **T: Think** — Think about the response from various perspectives and its potential impact.
- **S: Scrutinise** — Scrutinise the AI's conclusions by critically questioning and exploring alternative hypotheses or interpretations.

More info:

`https://teachersguidegsls.nl/genai/generative-ai-guidelines/`

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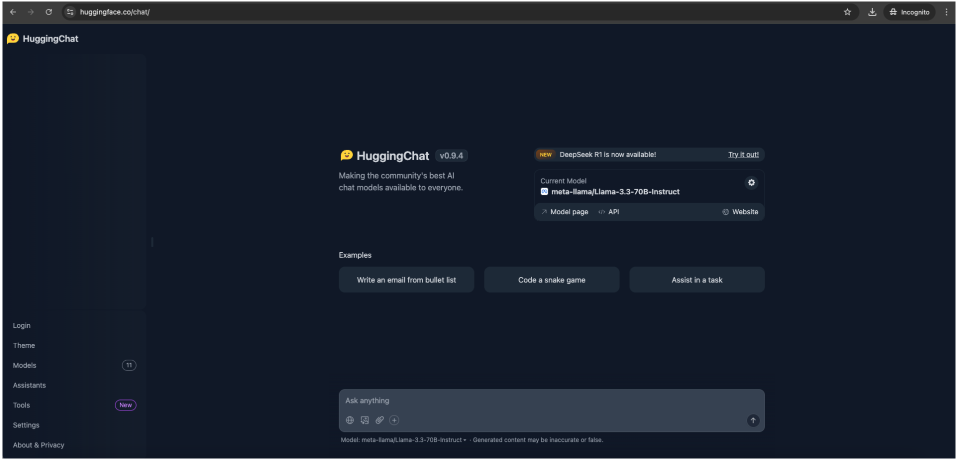
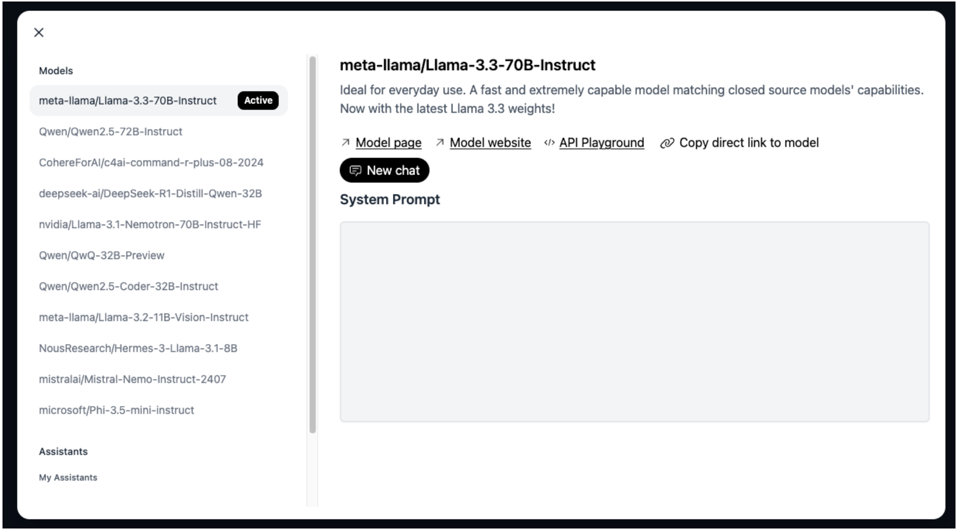
Ways to prompt



Switch models



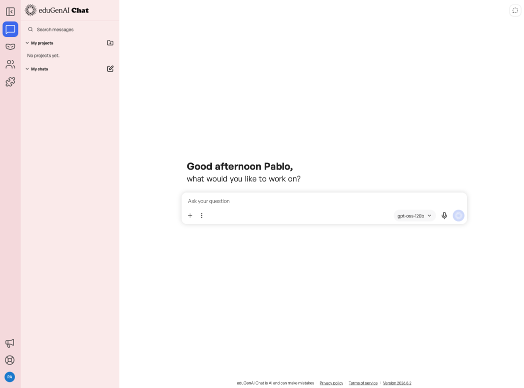
Official UU policy

`https://www.uu.nl/en/organisation/ai-policy/researchers`

- Benefits and use cases
- Risks and limitations
- Effective use of GenAI (*Background check*)
- Citing GenAI in your work

EduGenAI

`https://www.uu.nl/en/organisation/ai-policy/teachers/edugenai`



UU AI-chat

https://www.uu.nl/en/organisation/ai-policy/ai-platforms-at-uu

UU AI-chat

UU AI-Chat is Utrecht University's own generative AI chat environment. You use it via your browser to have conversations with various language models, within an environment managed by the University itself. The platform is based on the University of Amsterdam's AI chat environment. UU AI-Chat will be available in early September.

Who is it for?

It is intended for all students and staff at Utrecht University, for use in teaching, research and administrative operations. Guests and external parties cannot use the service.

What can you do with it?

More about UU AI Chat

How do you gain access?

This service will be available to students and staff from the start of September.

LLM Use in Research

Appropriate & Inappropriate Uses

- Think-Pair-Share

LLM Use in Research: Appropriate Applications

Appropriate Uses

- **Literature review:** Summarizing papers (with verification)
- **Writing assistance:** Drafting, editing, proofreading
- **Code generation:** Writing analysis scripts

> **Caveat**
>
> **only** when you know which statistical analysis is suitable

- **Brainstorming:** Generating research ideas
- **Translation:** Working with multilingual sources

Key: Always maintain human oversight and verification

LLM Use in Research: Inappropriate Applications

Avoid Using LLMs For:

- **Statistical analysis:** LLMs can't perform proper statistical tests
- **Fact-checking:** They may hallucinate information
- **Critical decisions:** Participant safety, ethical approvals
- **Original thinking:** Generating truly novel theories
- **Replacing expertise:** They supplement but don't replace researcher knowledge

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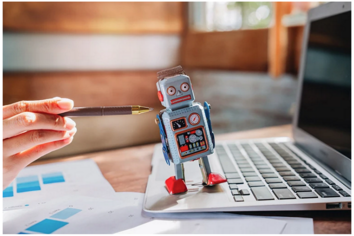

Using generative artificial intelligence models to kick ideas around is more effective than outsourcing thinking wholesale, says Yanjun Shen. Credit: jakkitticha/iStock via Getty

1. thinking for yourself
2. read literature
3. ask AI for advice
4. critically analyze

Key Takeaways: LLM Workflow

Remember:

- LLMs are powerful tools, not replacements for human expertise
- Always prioritize ethics, privacy, and transparency
- Verify outputs and maintain critical thinking